

CLINICAL TRIAL PROTOCOL

Title:

A Phase III Randomized Study Evaluating the Efficacy and Safety of LUMITAX versus Docetaxel in Patients with Metastatic Non-Small Cell Lung Cancer (NSCLC) After Progression on First-Line Chemo-Immunotherapy

Study Objectives:

Primary Objective:

- To demonstrate superiority of LUMITAX compared to docetaxel in Overall Survival (OS).

Secondary Objectives:

- To compare Progression-Free Survival (PFS) between treatment arms.
- To evaluate Patient Reported Outcomes (PROs) including quality of life (QoL).
- To assess Objective Response Rate (ORR) according to RECIST 1.1.
- To evaluate safety and tolerability profiles of both treatments.

Exploratory Objectives:

- To explore predictive biomarkers including PD-L1 expression and circulating tumor DNA (ctDNA).
- To evaluate immune-related biomarkers associated with response.

Study Design:

Multicenter, randomized (2:1), open-label Phase III study in patients with metastatic NSCLC who progressed after at least 4 cycles of platinum-based chemotherapy combined with anti-PD-1 or anti-PD-L1 therapy.

Patients will be randomized (2:1) into:

- Arm A (Experimental): LUMITAX monotherapy
- Arm B (Control): Docetaxel monotherapy

Stratification Factors:

- Histology (squamous vs non-squamous)
- ECOG Performance Status (0 vs 1)
- PD-L1 expression (<1% vs ≥1%)

Number of Patients:

Approximately 420 patients (280 in Arm A and 140 in Arm B).

Number of Sites:

Approximately 120 centers across Europe and North America.

Inclusion Criteria:

1. Age ≥ 18 years.
2. Histologically or cytologically confirmed metastatic NSCLC (Stage IV).
3. Documented progression after first-line platinum-based chemotherapy combined with anti-PD-1 or anti-PD-L1 therapy.
4. ECOG Performance Status 0-1.
5. At least one measurable lesion per RECIST 1.1.
6. Adequate organ function:
 - Absolute neutrophil count (ANC) ≥ $1.5 \times 10^9/L$
 - Platelets ≥ $100 \times 10^9/L$
 - Hemoglobin ≥ 9 g/dL
 - AST and ALT ≤ $2.5 \times ULN$ (≤ $5 \times ULN$ if liver metastases)
 - Total bilirubin ≤ $1.5 \times ULN$
 - Creatinine clearance ≥ 40 mL/min (CKD-EPI formula)
7. Women of childbearing potential must have a negative pregnancy test prior to treatment initiation.
8. Signed informed consent prior to any study-specific procedure.

Exclusion Criteria:

1. Known EGFR, ALK, or ROS1 genomic alterations with available approved targeted therapy.
2. Untreated or symptomatic brain metastases.
3. Active autoimmune disease requiring systemic immunosuppressive therapy.
4. Interstitial lung disease or active non-infectious pneumonitis.
5. Active infection requiring systemic therapy.
6. Prior exposure to LUMITAX.
7. Other active invasive malignancy within 3 years (except adequately treated non-melanoma skin cancer or carcinoma in situ).

Investigational Product:

Product Name: LUMITAX

Formulation: 10 mg/mL sterile solution

Route of Administration: Subcutaneous injection

Regimen: One injection every 3 weeks for 6 cycles, followed by maintenance dosing every 6 weeks.

Control Product:

Product Name: Docetaxel

Dose: 75 mg/m²

Route of Administration: Intravenous infusion over 1 hour

Regimen: One infusion every 3 weeks.

Duration of Treatment:

Treatment in both arms will continue until radiologic disease progression, unacceptable toxicity, withdrawal of consent, or for a maximum duration of 24 months.

Study Endpoints:

Primary Endpoint:

- Overall Survival (OS): defined as the time from randomization to death from any cause.

Secondary Endpoints:

- Progression-Free Survival (PFS)
- Objective Response Rate (ORR)
- Duration of Response (DoR)
- Change from baseline in Quality of Life scores (EORTC QLQ-C30)
- Time to ECOG Performance Status deterioration (>1)

Safety Endpoints:

- Incidence, severity, and causality of adverse events (AEs)
- Serious adverse events (SAEs)
- Immune-related adverse events (irAEs)
- Clinically significant laboratory abnormalities

Follow-Up:

After treatment discontinuation, patients will be followed every 12 weeks for survival status until death, loss to follow-up, or study closure.

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